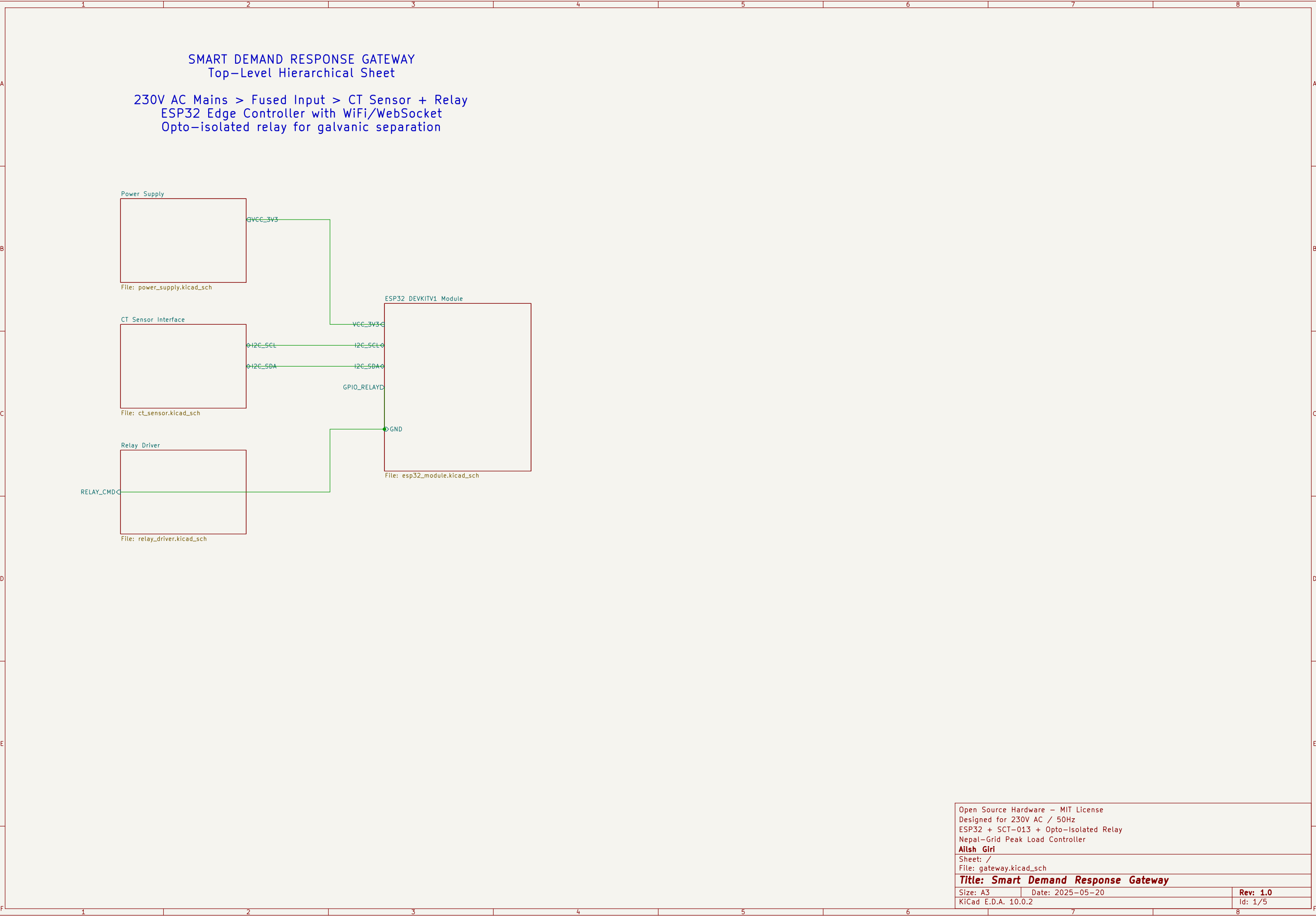
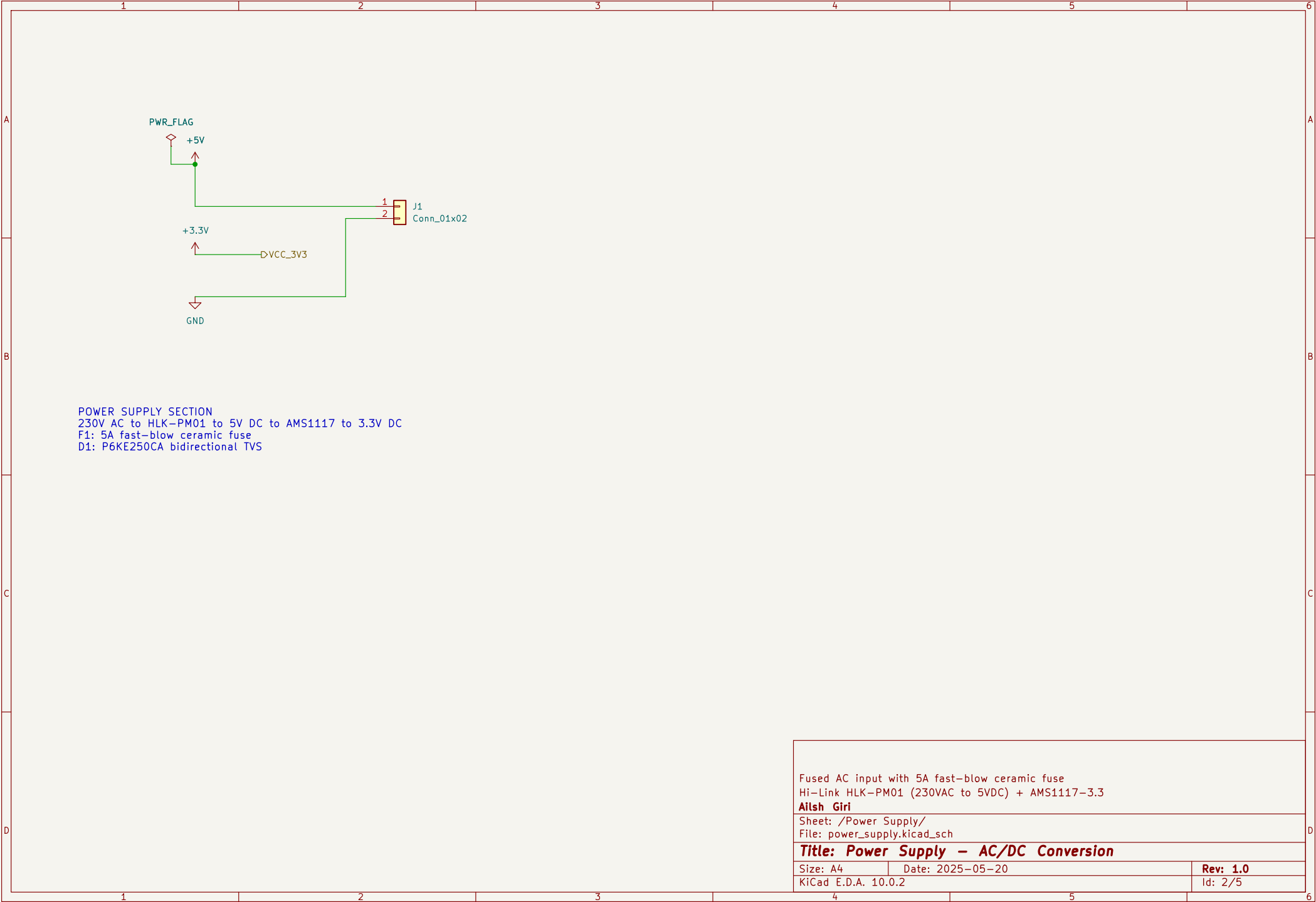
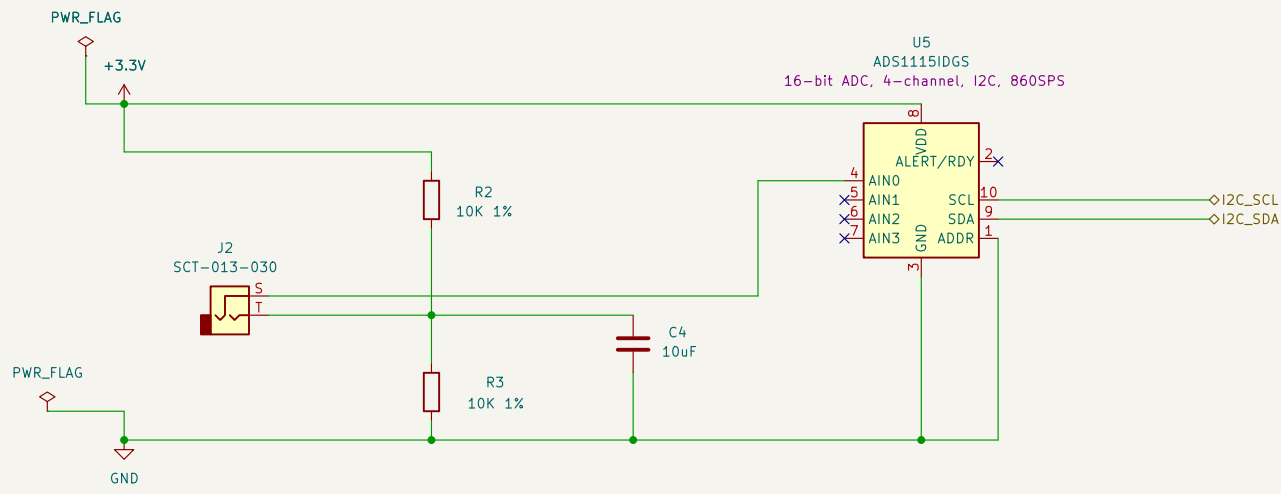








CT SENSOR INTERFACE WITH ADS1115
SCT-013 to Burden R (33 Ohm) to DC Bias Network to ADS1115 A0
R2/R3 divider (10K each) provides VCC/2 (1.65V) midpoint bias
C4 (10uF) filters bias midpoint noise
C5 AC-couples CT signal into biased ADC input
ADS1115 VDD=3.3V, ADDR=GND (I2C addr 0x48)
I2C SCL/SDA to ESP32 GPIO22/GPIO21
D2/D3 Schottky clamps limit ADS1115 input to 0V-3.3V

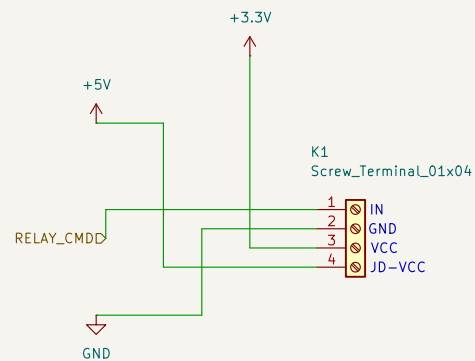
ADS1115 VDD=3.3V, ADDR=GND (0x48), I2C to ESP32
33 Ohm burden resistor, DC bias at VCC/2, ADS1115 16-bit ADC
Non-invasive AC current sensing

Ailsh Giri

Sheet: /CT Sensor Interface/
File: ct_sensor.kicad_sch

Title: CT Sensor Interface - SCT-013-030

Size: A4	Date: 2025-05-20	Rev: 1.0
KiCad E.D.A. 10.0.2		Id: 3/5



OPTO-ISOLATED RELAY DRIVER
ESP32 GPIO (3.3V) to R5 (330 Ohm) to PC817 LED to GND
PC817 phototransistor to R6 (1K) to Q1 Base
Q1 Collector to Relay Coil (SRD-05VDC) to VCC_5V
D4 flyback protection across coil
D5 TVS across relay contacts for arc suppression

Creepage greater than 6mm per IEC 60950-1
Galvanic isolation between 3.3V logic and 230V AC
PC817 optocoupler + NPN transistor + relay
Ailsh Giri

Sheet: /Relay Driver/
File: relay_driver.kicad_sch

Title: Relay Driver - Opto-Isolated Load Switching

Size: A4	Date: 2025-05-20	Rev: 1.0
KiCad E.D.A. 10.0.2		Id: 4/5

